

LAB MST

Que 1:

```
CREATE TABLE department1 (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(30)  
);
```

```
CREATE TABLE employee_4 (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    salary NUMERIC(10,2),  
    dept_id INT REFERENCES department1(dept_id)  
);
```

```
INSERT INTO department1 VALUES  
(1, 'IT'),  
(2, 'Sales'),  
(3, 'HR');
```

```
INSERT INTO employee_4 VALUES  
(101, 'Aman', 90000, 1),  
(102, 'Neha', 70000, 1),  
(103, 'Raj', 50000, 1),  
(104, 'Priya', 80000, 2),  
(105, 'Karan', 60000, 2),  
(106, 'Riya', 40000, 2),  
(107, 'Mohit', 75000, 3),  
(108, 'Simran', 55000, 3);
```

```
SELECT d.dept_name, e.emp_name, e.salary  
from employee_4 as e  
join department1 as d  
on e.dept_id=d.dept_id
```

```
where e.salary > (  
    select avg(salary)  
    from employee_4  
    where dept_id=e.dept_id  
)  
and e.dept_id in(
```

```

select dept_id
      from employee_4
      group by dept_id
      having count(*)>=3
)

```

	dept_name character varying (30)	emp_name character varying (50)	salary numeric (10,2)
1	IT	Aman	90000.00
2	Sales	Priya	80000.00

Que 2:

```

CREATE TABLE bank_customer (
  customer_id INT PRIMARY KEY,
  customer_name VARCHAR(50),
  balance NUMERIC(10,2)
);

```

```

CREATE TABLE customer_audit (
  audit_id SERIAL PRIMARY KEY,
  customer_id INT,
  customer_name VARCHAR(50),
  action VARCHAR(20),
  action_time TIMESTAMP
);

```

```

INSERT INTO bank_customer VALUES
(1, 'Rahul', 50000),
(2, 'Neha', 75000);

```

```

CREATE OR REPLACE FUNCTION log_customer_action()
returns trigger as $$

```

```

begin
IF TG_OP = 'INSERT' THEN
    INSERT INTO customer_audit
    (customer_id, customer_name, action, action_time)
    VALUES
    (NEW.customer_id, NEW.customer_name, 'ADDED',
CURRENT_TIMESTAMP);
    RETURN NEW;
ELSIF TG_OP = 'DELETE' THEN
    INSERT INTO customer_audit
    (customer_id, customer_name, action, action_time)
    VALUES
    (OLD.customer_id, OLD.customer_name, 'REMOVED',
CURRENT_TIMESTAMP);
    RETURN OLD;
END IF;
END;
$$ LANGUAGE plpgsql;

```

```

CREATE TRIGGER trg_customer_audit
AFTER INSERT OR DELETE
ON bank_customer
FOR EACH ROW
EXECUTE FUNCTION log_customer_action();

```

```

INSERT INTO bank_customer
VALUES (3, 'Aman', 60000);

```

```

DELETE FROM bank_customer
WHERE customer_id = 1;

```

```

select * from customer_audit;

```

	audit_id [PK] integer	customer_id integer	customer_name character varying (50)	action character varying (20)	action_time timestamp without time zone
1	1	3	Aman	ADDED	2026-09-28 10:23:49.108313
2	2	1	Rahul	REMOVED	2026-09-28 10:23:49.108313